

ROCHESTER PIER, NY, USA

WMO: 998009

Lat: 43.2625N

Lon: 77.5981W

Elev: 272

StdP: 14.55

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.3	(c) 10.6	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 32.2	(k) 25.2	(l) 28.9	(m) 22.2	(n) 14.6	(o) 210	(p) 0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.5	(c) 86.0	(d) N/A	(e) 82.7	(f) N/A	(g) 80.0	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 12.4	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 27.9	(b) 24.9	(c) 22.3	(e) 3.2	(f) 90.9	(g) 4.4	(h) 1.8	(i) 0.1	(j) 92.2	(k) -2.5	(l) 93.3	(m) -4.9	(n) 94.3	(o) -8.1	(p) 95.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	26.9	27.9	34.8	46.2	57.4	65.3	72.2	70.5	64.6	53.5	42.6	34.0		
	DBStd	17.64	10.57	9.77	8.96	7.91	7.77	6.68	5.19	4.30	6.61	6.83	8.53	8.55		
	HDD50	2800	722	623	478	168	16	0	0	0	0	41	251	501		
	HDD65	6155	1182	1040	936	567	263	78	4	5	85	362	672	961		
	CDD50	2717	4	3	7	54	245	458	688	636	439	149	29	5		
	CDD65	597	0	0	0	3	27	86	227	175	74	5	0	0		
	CDH74	2832	0	0	0	26	173	440	1174	640	356	23	0	0		
	CDH80	608	0	0	0	5	35	112	260	103	89	4	0	0		
(14) Wind	WSAvg	10.8	12.7	12.8	12.5	11.9	9.1	8.6	8.5	9.0	9.6	11.4	11.5	11.9		
(15) Precipitation	PrecAvg	33.2	2.3	2.0	2.4	3.1	3.1	2.9	3.3	3.1	3.0	3.2	2.5	2.4		
	PrecMax	42.1	5.7	4.0	4.8	5.8	5.8	5.8	7.3	5.1	5.2	6.0	3.9	4.5		
	PrecMin	26.9	1.0	0.6	0.8	1.0	0.8	0.3	1.1	1.0	1.2	1.1	0.6	1.1		
	PrecStd	3.6	1.1	0.8	1.0	1.2	1.3	1.6	1.8	1.0	1.1	1.4	0.9	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.9	61.1	64.3	79.7	84.5	88.3	90.5	87.1	88.1	76.7	68.2	62.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	53.9	52.8	59.3	70.8	78.7	83.3	86.0	82.8	82.0	70.4	63.7	55.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	46.7	45.9	53.3	65.2	73.9	79.3	82.6	79.7	77.5	66.2	59.3	50.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	40.9	39.5	47.0	59.5	69.2	75.1	80.0	77.4	74.5	63.4	55.0	46.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	11.1	11.2	13.3	14.2	11.4	10.5	10.5	11.5	11.1	10.8	9.5		
		MCDBR	18.5	20.7	20.8	25.1	20.7	16.7	13.8	13.7	15.3	16.0	16.9	15.4		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.309	0.317	0.335	0.373	0.397	0.439	0.435	0.423	0.378	0.360	0.316	0.296		
	taud		2.396	2.370	2.394	2.344	2.329	2.248	2.270	2.288	2.397	2.428	2.511	2.491		
	Ebn at Noon		260	279	287	283	278	266	265	264	269	259	258	257		
	Edh at Noon		25	30	33	38	39	43	42	40	33	28	22	21		
(44) All-Sky Solar Radiation	RadAvg		407	655	1065	1429	1750	1859	1884	1673	1330	798	490	321		
	RadStd		38	62	100	153	177	150	144	93	110	74	65	48		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page